
ABSTRACT LINEAR ALGEBRA FINAL EXAM: TERMS AND RESULTS

TEN IMPORTANT TERMS

A **vector space** is a set V of vectors with the structure of a commutative group under addition, a field $\mathbb{F}$ of scalars, and a multiplication of vectors by scalars that distributes over addition of vectors.

A **linear combination** of vectors is a finite sum of scalar multiples of the vectors. For vectors $v_1, v_2, \dots, v_p$, a linear combination of is a sum of form $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_p v_p = \sum_1^p \alpha_k v_k$ where the α_k are scalars. The **span** of a set of vectors in a vector space is the set of all of their linear combinations.

A **linear transformation** is a function between vector spaces over the same field that respects the operations of vector addition and scalar multiplication: $L(\alpha v + \beta w) = \alpha L(v) + \beta L(w)$.

A **basis** of a vector space V is a set of vectors in V such that any vector v in V admits a unique representation as a linear combination of the vectors in the basis. The cardinality of a basis is the **dimension** of a vector space. (*Note: it is a theorem that dimension is well-defined.*)

A matrix is in **row echelon form** if all zero rows (if any) are below all non-zero rows, and if every non-zero row has its leading non-zero entry strictly to the right of the first non-zero entry in the previous row; the leading non-zero entry in each non-zero row is called its **pivot**. A matrix is in **reduced row echelon form** if it also has a 1 for each of its pivots and all entries above each pivot are 0.

The **kernel** of a linear transformation $L : V \rightarrow W$ is the subset of V sent to the zero vector in W by L . The **range** of L is the subset of W in the image of L . (*It is a result that the kernel and range are subspaces of V and W , respectively; they are also called the **null space** and **column space**.*)

Two matrices A and B are **similar** (or **conjugate**) if there exists an invertible matrix Q such that $A = Q^{-1}BQ$.

For the square matrices $M_{n \times n}$ over a field $\mathbb{F}$, a **determinant** is a function $\det : M_{n \times n} \rightarrow \mathbb{F}$ such that $\det$ is multilinear and alternating and $\det(I) = 1$, where I is the $n \times n$ identity matrix. (*Note: it is a theorem that a determinant exists and is unique.*)

A non-zero vector v is an **eigenvector** of an operator A if there exists a scalar λ such that $Av = \lambda v$. The scalar λ is the **eigenvalue** for the eigenvector v .

In an inner product space, a set of vectors is **orthonormal** if each vector has norm 1 and each pair of distinct vectors is orthogonal (so that the inner product of the pair is 0).

TEN IMPORTANT RESULTS: CHAPTERS 1-3

Any finite generating system for a vector space contains a basis. Every basis of a vector space has the same cardinality, yielding a well-defined dimension. (Prop. 1.2.8; Prop. 2.3.3)

Among the set of matrices that are row equivalent to a given matrix, there exists a unique matrix in reduced row echelon form. (Exercise 2.3.9; see handout.)

The solution set of a linear system $Ax = b$ is a translate of the solution set of $Ax = 0$ by a specific solution. (Thm. 2.6.1)

The Invertible Matrix Theorem. Let A be a square $n \times n$ matrix with entries in $\mathbb{F}$, and let $T : V \rightarrow V$ be the linear transformation $T(x) = Ax$. The following statements are equivalent.

- (1) A is invertible.
- (2) T is invertible.
- (3) A is row equivalent to I_n .
- (4) A has n pivots. (Prop. 2.3.6)
- (5a) $Ax = b$ has a unique solution for all b in $\mathbb{F}^n$. (Thm. 1.6.8)
- (5b) T is a bijection.
- (5c) The columns of A form a basis for $\mathbb{F}^n$. (Cor. 1.6.9)
- (6a) $Ax = 0$ has only the trivial solution. ($\text{Nul } A = \{0\}$)
- (6b) T is one-to-one.
- (6c) The columns of A are linearly independent.
- (7a) $Ax = b$ is consistent for all b in $\mathbb{F}^n$.
- (7b) T is onto.
- (7c) The columns of A span $\mathbb{F}^n$. ($\text{Col } A = \mathbb{F}^n$)
- (8) $\det(A) \neq 0$.
- (9) $0 \in \mathbb{F}$ is not an eigenvalue of A . (*not until Chapter 4*)

Rank Theorem (Thm. 2.7.1). For a matrix A , $\text{rank}(A) = \text{rank}(A^T)$.

Rank-Nullity Theorem (Thm. 2.7.2): For an $m \times n$ matrix A , $\text{rank}(A) + \dim(\text{Nul}(A)) = n$.

For any vector space of matrices, a determinant function exists and is unique.

Cofactor expansion of determinant (Thm 3.5.1) For an $n \times n$ matrix A , let $A_{j,k}$ denote the $(n-1) \times (n-1)$ matrix obtained from A by crossing out row number j and column number k . For any fixed j , we have

$$\det A = \sum_{k=1}^n a_{j,k} (-1)^{j+k} \det A_{j,k}.$$

Cramer's rule: For an invertible matrix A , the k th entry x_k of the solution of the equation $Ax = b$ is given by the formula

$$x_k = \frac{\det B_k}{\det A},$$

where the matrix B_k is obtained from A by replacing column number k of A by the vector b .

For $n \times n$ matrices A and B , $\det(AB) = \det(A) \det(B)$ and $\det(A^{-1}) = \det(A)^{-1}$ (Thm. 3.3.5)

FIVE IMPORTANT RESULTS: CHAPTERS 4-6 & 9

(Thm. 4.1.2) For a square complex matrix A , the product of the eigenvalues of A equals $\det(A)$ and the sum of the eigenvalues of A equals $\operatorname{tr}(A)$.

Cauchy–Schwarz inequality (Theorem 5.1.7) Let $|\cdot|$ be the usual absolute value for real numbers and let $\cdot$ be ordinary multiplication of real numbers. For vectors x and y in an inner product space with inner product $(\cdot, \cdot)$ and norm $\|\cdot\|$, $|(x, y)| \leq \|x\| \cdot \|y\|$.

(Gram–Schmidt process) Every subspace of an inner product space has an orthonormal basis.

Cayley–Hamilton Theorem (Thm. 9.1.1). Let A be a square matrix, and let $p(\lambda) = \det(A - \lambda I)$ be its characteristic polynomial. Then $p(A) = 0$, where 0 is the zero matrix.

The minimal polynomial of a square matrix divides its characteristic polynomial.

TEN IMPORTANT MATRIX DECOMPOSITIONS

Product of elementary matrices: Every invertible matrix can be written as a product of elementary matrices. (Prop. 2.4.1 & Lemma 3.3.8) [row reduction]

PLU and LU decompositions: Let A be a square matrix. Then $A = PLU$, where P is a permutation matrix, L is a lower triangular matrix, and U is an upper triangular matrix. If A can be put in row echelon form without permuting rows or adding a multiple of a higher row to a lower row, then A admits an LU decomposition. [row reduction]

Matrix equivalence: Every $m \times n$ matrix A can be written as $A = P“I”Q$, where P and Q are invertible $m \times m$ and $n \times n$ matrices and “ I ” is an “identity” matrix with 1’s on the beginning of the diagonal and 0’s elsewhere. [row and column reduction, matrix equivalence]

Diagonalization (Thm 4.2.1): A matrix A (with values in $\mathbb{F}$) admits a representation $A = SDS^{-1}$, where D is a diagonal matrix and S is an invertible one (both with entries in $\mathbb{F}$) if and only if there exists a basis in $\mathbb{F}^n$ of eigenvectors of A . Moreover, in this case the diagonal entries of D are the eigenvalues of A and the columns of S are the corresponding eigenvectors (column number k corresponds to k th diagonal entry of D).

QR decomposition: Let A be an invertible matrix in an inner product space. Then $A = QR$ where Q has orthonormal columns and R is upper triangular with positive diagonal entries. [Gram–Schmidt]

Upper triangular (Schur) representation (Thms. 6.1.1 & 6.1.2): Let A be the matrix of an operator on an inner product space. Then A can be represented as $A = UTU^*$, where U is a unitary matrix and T is an upper triangular matrix. In the case of real square matrices, any real square matrix A with all real eigenvalues can be represented as $A = UTU^{-1}$, where U is a real orthogonal matrix and T is a real upper triangular matrix.

Spectral Theorem (Theorem 6.2.1): Let $A = A^*$ be a self-adjoint operator in an inner product space X (the space can be complex or real). Then all eigenvalues of A are real, and there exists an orthonormal basis of eigenvectors of A in X . In the real case, this says that symmetric matrices ($A = A^T$) are diagonalizable with an orthonormal change of basis matrix: $A = QDQ^{-1}$

Singular Value Decomposition (Def 6.3.9, see Kalman article for proof): Every matrix A has a singular value decomposition $A = W\Sigma V^*$, where W and V are unitary (orthogonal for $\mathbb{R}$) matrices and Σ is “diagonal” with real numbers on the beginning of the diagonal and 0’s elsewhere.

D+N (Cor. 9.3.9; Jordan–Chevalley decomposition): Any operator A in V can be represented as $A = D + N$, where D is diagonalizable (i.e. diagonal in some basis) and N is nilpotent ($N^m = 0$ for some m), and $DN = ND$.

Jordan decomposition theorem (Thm. 9.5.1): Given an operator T there exist a basis (Jordan canonical basis) such that the matrix A of T in this basis has a block diagonal form with blocks of the form

$$\begin{pmatrix} \lambda & 1 & & 0 \\ & \lambda & 1 & \\ & & \lambda & \ddots \\ & & & \ddots & 1 \\ 0 & & & & \lambda \end{pmatrix}$$

where λ is an eigenvalue of A . (Here we assume that the block of size 1 is just λ .) The matrix A is unique up to permutation of the blocks.

Chapters 1 and 2:

vector
scalar
group
field
vector space
linear transformation
linear operator
matrix
subspace
transpose, conjugate transpose
linear combination
trivial (linear combination, solution, subspace)
linear independence (and dependence)
span (generate)
spanning system
basis
standard basis
coordinates (in a basis)
dimension
isomorphism
rank
column space (range)
null space (kernel)
row space
left null space
one-to-one (injective)
onto (surjective)
isomorphism
operator
similarity, conjugation, change of basis
homogeneous and inhomogeneous systems
vector equation
matrix equation
inconsistent and consistent systems
unique solution
trivial solution
general solution

coefficient matrix
augmented matrix
row operations
row equivalent / column equivalent
row reduction / column reduction
(row) echelon form
reduced row echelon form
pivot
free variable
(parametric) vector form
fundamental subspaces of a matrix
matrix equivalent
trace

types of matrices/linear transformations:

zero
identity
elementary
nilpotent
projection
scaling
shear
rotation
permutation
periodic (finite-order)
orthogonal (isometry)
unitary
diagonal, block diagonal
(lower / upper) triangular
invertible (nonsingular)
non-invertible (singular)
Jordan Form
symmetric (equal to transpose)
self-adjoint (equal to conjugate transpose, Hermitian)
normal (commutes with its adjoint)

Chapter 3:

determinant
parallelepiped
submatrix

minor of order k

cofactor expansion

multilinear

antisymmetric, alternating

Chapter 4:

eigenvalue

eigenvector

eigenspace

characteristic polynomial

invariant subspace

algebraic multiplicity

geometric multiplicity

spectrum

diagonalization

Chapters 5 & 6

inner product

inner product space

norm

normed space

dot product

orthogonal vectors

orthonormal basis

normalized

orthogonal complement

singular value

Chapter 9

minimal polynomial

generalized eigenvector

generalized eigenspace

Jordan block